

PixelPaws documentation

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vercel links: <https://pixelpaws-git-main-shenyee-s-projects.vercel.app/>

video demo links: [PixelPaws-20260616_002656-Meeting Recording.mp4](#)

1. How the App Was Planned and Approached

The development of this study lounge prototype is approached using an Iterative, Component-Driven Design strategy. Because the app bridges the gap between a standard web utility (to-do lists, timers) and a 2D game environment (Phaser.js canvas), the architecture is split into two primary layers:

- The Web UI Layer (DOM): Built using modern web frameworks to handle highly interactive, data-driven interfaces like the collapsible sidebar, to-do tracking, registration systems, and user authentication.
- The Game Engine Layer (Canvas): Built inside Phaser.js to render pixel-crisp sprites, manage tilemap collisions, animate characters/pets, and process game logic.

The application is built in three logical phases to ensure scalability and performance:

- Phase A (State Centered Core): A single source of truth (state.js) handles the game loop (XP, coins, pet health, goals). This ensures UI updates instantly regardless of the current page view.
- Phase B (Separated Presentation Layers): Low coupling is achieved by separating page logic into distinct modules, such as app.js and goals.js for the Home dashboard, and lounge.js for the interactive canvas.
- Phase C (Reactive Syncing): An event emitter pattern (state.on / state.trigger) binds UI elements to state changes, allowing instant visual feedback (like flying hearts) when a task is checked.

2. Problem Statement:

Current tools for managing tasks and productivity tend to come off as cold, clinical, or mechanical in their use, which leads to fast user burnout and dropped habits. Similarly, commonly used habit tracking apps (with gamification) often don't provide sufficient stakes or immediate visually compelling results to keep users accountable.

For students and remote workers, an environment can be created to provide immediate and emotionally compelling anchors by taking away feelings of existential separation from studying. It is also difficult to maintain long-term engagement with self-reported task lists without both an actualized way to measure progress and an incentive to stay engaged (retention mechanics).

3. Project Goals:

Create an interactive, top-down 2D pixel-art style digital study lounge prototype that combines structural productivity items/units with an emotional connection item/units virtual pet ecosystem.

By connecting task completion with a virtual pet's health, survival, and visible level you create the ability to hold users accountable for what they did during their study sessions in an engaging way, while creating an interactive cafe/virtual pet park experience.

4. Tech Stack & Tooling

- Frontend Core: Vanilla HTML5, CSS3 (using Custom Properties for theming), and ES6 JavaScript Modules.
- Development Tool: Vite 5.x for ultra-fast Hot Module Replacement (HMR) and optimized asset bundling.
- Database & Auth: Firebase Auth and Cloud Firestore for seamless cloud syncing.
- Assets: Custom pixel art (PNGs) and inline SVG vectors.

5. Main Technical Decisions

Event-Driven Communication: Utilizing a pub/sub system prevents the application from crashing if a triggered event lacks an active listener on the current page.

Decoupled Pet Stat Loop: Pet decay runs on background intervals, dynamically adjusting based on active study timers to gamify the experience.

Event Delegation: Binding listeners to static parent containers prevents memory leaks when dynamic elements are rendered or destroyed.

CSS Variables: Allows for instant theming (like Dark Mode) without rewriting structural CSS.

6. Flowchart:

Since the flowchart is too big, please refer to the links below to check it out.

https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&target=blank&highlight=0000ff&edit=_blank&layers=1&nav=1&dark=0#G1ctiNv4RiUzYedzKyHmIDy24iS6BYsMP_